VISUAL INSPECTION PROCEDURE

(Doc No. CT8-PRO-CM-012)

USA PROJECT UNIT 1 - TALLGRASS

Revision History				
Revisions	*Date*	*Descriptions*	*Prepared By*	*Approved By*
0	25-Mar-26	First Issue	Ms. Michelle Wilrutstien	Mr. Phisit Pew-la-or

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Signature:					

Table of Contents

1. SCOPE 3

2. REFERENCE DOCUMENTS 3

3. NDT PERSONNEL QUALIFICATION 3

4. EQUIPMENT AND MATERIALS 3

5. PREPARATION 3

6. PROCEDURES 4

7. ACCEPTANCE CRITERIA 5

8. REPORTING 5

9. APPENDIXES 6

1. SCOPE

This procedure prescribes the minimum requirements for visual inspection is performed in accordance with the project and code requirements for all welds of structural modularization for USA PROJECT UNIT 1 - TALLGRASS.

2. REFERENCE DOCUMENTS

2.1 Code and Standard

- ASME B31.1 : Power Piping Code
- ISO 5817 : Welding – Fusion-Welded joints in steel, nickel, titanium and their alloys (beam welding excluded) – Quality levels for imperfections.
- AWS D1.1 : Structural Welding Code

2.2 Customer and Client Specifications

- LU-00130 : Quality Requirements for Project

3. NDT PERSONNEL QUALIFICATION

Personnel who performing visual examinations shall be experienced and qualified per the referenced international Standards and with relevant visual inspection certification schemes such as the CSWIP or AWS.

4. EQUIPMENT AND MATERIALS

As a minimum inspection mirrors, magnifying, glasses, welding gauges, fillet weld gauge, tape measures, ruler, flashlight, chalk/crayons and cameras shall be used in performing visual examination.

5. PREPARATION

5.1 Direct visual examination the surface to be examined shall be viewed from a distance not to exceed 24 inches and at angle of not less than 30 degree to the surface to be examined. All surfaces to be examined shall be adequately illuminated by either natural or artificial means. All surfaces to be examined shall be clean and free of irregularities which can mask or contused with the defect.

5.2 Remote Visual Examination, remote techniques may be employed when difficult access makes direct examination impractical. Remote visual examinations may be performed by use of mirrors, telescopes, borescopes, fiber optics, cameras, or any other suitable instruments. When remote systems are necessary, the apparatus shall have a demonstrated resolution capability equivalent to that obtained by direct visual examination.

5.3 Required Illumination, the surface of the part undergoing examination maybe be illuminated by any means that will provide a minimum of 100-foot candles (1000 lux).

6. PROCEDURES

6.1 The Inspector shall make certain that only materials conforming to the project specifications requirements, approved drawings shall be used.

6.2 Material shall be pre-heated according to the project specific requirements if welding seam preparations are done by thermal cutting. The thermal cutting shall be followed by machining to achieve the final weld joint end detail. End preparation is acceptable only if the surface is reasonably smooth and true, and free of slag. Discoloration remaining on a thermally cut surface shall be removed

6.3 When fit-up examinations of butt joint are required, it shall be checked for proper alignment, root opening and root face on the edge preparation of the joint shall meet the WPS requirement. The surface to be joined shall be checked to ensure full penetration and smooth transition into the adjacent base material.

6.4 Joint geometry shall be in accordance with the Welding Procedure Specification (WPS).

6.5 Where accessible, the inside surface of butt joint shall be examined for misalignment.

6.6 Weld alignment and fit-up inspection shall be performed 100% of all welds.

6.7 For socket welding, gap between pipe and flange or fitting shall be as per the project or code requirements.

6.8 Tack welds Piping (if application) - The welding material shall be the same as specified in the welding procedure. The welding works shall be performed by qualified and approved welders, bridge tacks (above the weld) shall be removed. The removal shall not reduce the base metal thickness allowed by the code/standard and the area around is marked for liquid penetrant or magnetic test examination.

6.9 Tack weld Structure - Cracked or oversized tack welds shall be removed prior to depositing the final weld. Fillet type welds shall be checked for proper alignment, condition of tack welds, cleanliness and base metal suitability. Tack weld shall be made by qualified welder using approved welding procedure.

6.10 Inspection during welding - Welding Inspector shall be monitoring welding parameters in accordance with approved welding procedures and record detail of weld procedure variables, such currents, voltage, travel speed during welders performing welding process with calibrated measuring and test equipment.

Where back gouging is required, the inspector shall inspect the back gouged area to ensure that the gouged area is free from any carbon contamination and is of clean sound metal and ensure the surface after gouged have been prepared in accordance with the joint details as per the drawings and applicable weld procedure specification prior to welding.

During the welding process the inspector shall monitor to ensure that the cleaning between each pass is free from possible inclusions which may cause weld defects.

6.11 Inspection after welding

6.11.1 Complete welded

- The welds shall be examined for irregularities and defects. All completed welds shall be 100% visually inspected.
- Weld metal reinforcements and weld profiles shall be checked to ensure it is within the limits of the applicable codes/standards and project standard details drawings.
- Fillet welds shall be checked with a suitable gauge to the design dimensions specified.

6.11.2 Check Dimension

- All dimensions shall be in accordance with shop drawing comparison with design drawing.

6.11.3 Final Inspection

- Verification with shop drawing, dimensional report, painting report, NDT report compliant with contract documents.

7. ACCEPTANCE CRITERIA

The following acceptance standards for weld shall be applied as appropriate except where requirements are specified for specific materials or applications.

- ASME B31.1 : 2018 – Para. 136.4.2 and Table 127.4.2-1
- ISO 5817 : 2014 – Table 1 – Limits for imperfections, Item 1, 3 & 4
- AWS D1.1 : Structural Welding Code

8. REPORTING

Any imperfection found during examinations shall be recorded into AFC DWGs for further traceability and follow-up.

9. APPENDIXES

APPENDIX 1 - Reference ASME B31.1 Acceptance criteria

136.3 Examination

136.3.1 General. Examination denotes the functions performed by the manufacturer, fabricator, erector, or a party authorized by the owner that include nondestructive examinations (NDE), such as visual, radiography, ultrasonic, eddy current, liquid penetrant, and magnetic particle methods. The degree of examination and the acceptance standards beyond the requirements of this Code shall be a matter of prior agreement between the manufacturer, fabricator, or erector and the owner.

(18) **136.3.2 Qualification of NDE Personnel.** Personnel who perform nondestructive examination of welds shall be qualified and certified for each examination method in accordance with a program established by the employer of the personnel being certified, which shall be based on the following minimum requirements:

(a) instruction in the fundamentals of the nondestructive examination method.

(b) on-the-job training to familiarize the NDE personnel with the appearance and interpretation of indications of weld defects. The length of time for such training shall be sufficient to ensure adequate assimilation of the knowledge required.

(c) an eye examination performed at least once each year to determine optical capability of NDE personnel to perform the required examinations.

(d) upon completion of (a) and (b), the NDE personnel shall be given a written examination and performance examination by the employer to determine if the NDE personnel are qualified to perform the required examinations and interpretation of results.

(e) certified NDE personnel whose work has not included performance of a specific examination method for a period of 1 yr or more shall be recertified by successfully completing the examination of (d) and also passing the visual examination of (c). Substantial changes in procedures or equipment shall require recertification of the NDE personnel.

As an alternative to the preceding program, the requirements of ASME BPVC, Section V, Article 1, T-120(e) or T-120(f) may be used for the qualification of NDE personnel. Personnel qualified to AWS QC1 may be used for the visual examination of welds provided they meet the annual eye examination requirement of (c) and the J1 visual acuity requirement of ASME BPVC, Section V, Article 9.

136.4 Examination Methods of Welds

136.4.1 Nondestructive Examination. Nondestructive examinations shall be performed in accordance with the requirements of this Chapter. The types and extent of mandatory examinations for pressure welds and welds to pressure retaining components are specified in Table 136.4.1-1. For welds other than those covered by Table 136.4.1-1, only visual examination is required. Welds requiring nondestructive examination shall comply with the applicable acceptance standards for indications as specified in paras. 136.4.2 through 136.4.6. As a guide, the detection capabilities for the examination method are shown in Table 136.4.1-2. Welds not requiring examination (i.e., RT, UT, MT, or PT) by this Code or the engineering design shall be judged acceptable if they meet the examination requirements of para. 136.4.2 and the pressure test requirements specified in para. 137. NDE for P-Nos. 3, 4, 5A, 5B, and 15E material welds shall be performed after postweld heat treatment unless directed otherwise by engineering design. Required NDE for welds in all other materials may be performed before or after postweld heat treatment.

136.4.2 Visual Examination. Visual examination as defined in para. 100.2 shall be performed in accordance with the requirements described in ASME BPVC, Section V, Article 9. Visual examinations may be conducted, as necessary, during the fabrication and erection of piping components to provide verification that the design and WPS requirements are being met. In addition, visual examination shall be performed to verify that all completed welds in pipe and piping components comply with the acceptance standards specified in (a) or with the limitations on imperfections specified in the material specification under which the pipe or component was furnished.

(a) Acceptance Standards. The following indications are unacceptable:

(1) cracks — external surface.

(2) undercut on the surface that is greater than $^1/_{32}$ in. (1.0 mm) deep, or encroaches on the minimum required section thickness.

(3) undercut on the surface of longitudinal butt welds.

(4) weld reinforcement greater than specified in Table 127.4.2-1.

(5) lack of fusion on surface.

(6) incomplete penetration (applies only when inside surface is readily accessible).

(7) any other linear indications greater than $^3/_{16}$ in. (5.0 mm) long.

(8) surface porosity with rounded indications having dimensions greater than $^3/_{16}$ in. (5.0 mm) or four or more rounded indications separated by $^1/_{16}$ in. (2.0 mm) or less edge to edge in any direction. Rounded indications are indications that are circular or elliptical with their length less than three times their width.

136.4.3 Magnetic Particle Examination. Whenever required by this Chapter (see Table 136.4.1-1), magnetic particle examination shall be performed in accordance with the requirements of ASME BPVC, Section V, Article 7.

(a) Evaluation of Indications

(1) Mechanical discontinuities at the surface will be indicated by the retention of the examination medium. All indications are not necessarily defects; however, certain

(18)

Table 127.4.2-1 Reinforcement of Girth and Longitudinal Butt Welds

Thickness of Base Metal, in. (mm)	Maximum Thickness of Reinforcement for Design Temperature >750°F (400°C)		≤750°F (400°C)	
	in.	mm	in.	mm
Up to 1/8 (3.0), incl.	1/16	2.0	3/32	2.5
Over 1/8 to 3/16 (3.0 to 5.0), incl.	1/16	2.0	1/8	3.0
Over 3/16 to 1/2 (5.0 to 13.0), incl.	1/16	2.0	5/32	4.0
Over 1/2 to 1 (13.0 to 25.0), incl.	3/32	2.5	3/16	5.0
Over 1 to 2 (25.0 to 50.0), incl.	1/8	3.0	1/4	6.0
Over 2 (50.0)	5/32	4.0	See Note (1)	See Note (2)

GENERAL NOTES:

(a) For double-sided groove welds, this limitation on reinforcement given above shall apply separately to both inside and outside surfaces of the joint.

(b) For single-sided groove welds with backing strips or bars that remain in place, the limitation on reinforcement given above shall apply to the outside surface. For single-sided groove welds without backing strips or bars that remain in place, the limits shall apply to the outside surface; they also apply to the inside surface when the inside surface is readily accessible.

(c) The thickness of weld reinforcement shall be based on the thickness of the thinner of the materials being joined.

(d) The weld reinforcement thicknesses shall be determined from the higher of the abutting surfaces involved.

(e) Weld reinforcement may be removed if so desired.

NOTES:

(1) The greater of 1/4 in. or 1/8 times the width of the weld in inches.

(2) The greater of 6 mm or 1/8 times the width of the weld in millimeters.

The full penetration groove welds shall be finished with cover fillet welds and meet the requirements of para. 104. The cover fillet welds shall have a minimum throat dimension not less than that shown in Figure 127.4.8-5 or Figure 127.4.8-6, as applicable.

(d) In branch connections having reinforcement pads or saddles, the reinforcement shall be attached by welds at the outer edge and at the branch periphery as follows:

(1) If the weld joining the added reinforcement to the branch is a full penetration groove weld, it shall be finished with a cover fillet weld having a minimum throat dimension not less than t_c; the weld at the outer edge, joining the added reinforcement to the run, shall be a fillet weld with a minimum throat dimension of $0.5t_{nr}$.

(2) If the weld joining the added reinforcement to the branch is a fillet weld, the throat dimension shall not be less than $0.7t_{min}$. The weld at the outer edge joining the outer reinforcement to the run shall also be a fillet weld with a minimum throat dimension of $0.5t_{nr}$.

(e) When rings or saddles are used, a vent hole shall be provided (at the side and not at the crotch) in the ring or saddle to reveal leakage in the weld between branch and main run and to provide venting during welding and heat treating operations. Rings or saddles may be made in more than one piece if the joints between the pieces have strength equivalent to ring or saddle parent metal and if each piece is provided with a vent hole. A good fit shall be provided between reinforcing rings or saddles and the parts to which they are attached.

(f) Branch connections NPS 2 (DN 50) and smaller that do not require reinforcements (see para. 104.3) may be constructed as shown in Figure 127.4.8-7. The groove welds shall be finished with cover fillet welds with a minimum throat dimension not less than that shown in Figure 127.4.8-7. This construction shall not be used at design temperatures greater than 750°F (400°C) nor at design pressures greater than 1,025 psi (7 100 kPa).

127.4.9 Attachment Welds. Structural attachments may be made by complete penetration, partial penetration, or fillet welds.

Low-energy capacitor discharge welding may be used for welding temporary attachments (e.g., thermocouples) and permanent nonstructural attachments without preheat above 50°F (10°C) or subsequent postweld heat treatment on P-No. 1 through P-No. 5B and P-No. 15E materials, provided that the following requirements are met:

(a) A Welding Procedure Specification is prepared, describing the low-energy capacitor discharge equipment, the combination of materials to be joined, and the technique of application; qualification of the welding procedure is not required.

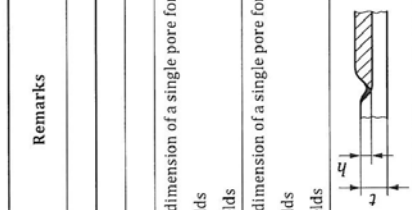
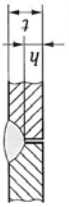
(b) The energy output of the welding process is limited to 125 W-sec.

(c) For P-No. 5A, P-No. 5B, and P-No. 15E materials, the maximum carbon content of the material is 0.15%.

(d) After temporary attachments are removed, the areas shall be examined in accordance with para. 136.4.

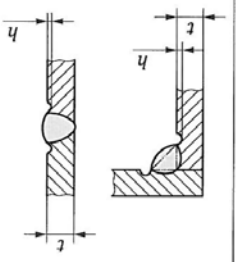
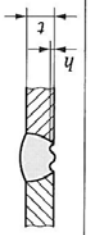
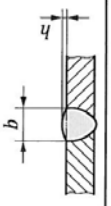
APPENDIX 2 - Reference ISO 5817 Acceptance criteria

Table 1 — Limits for imperfections

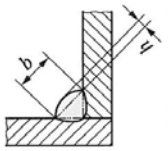
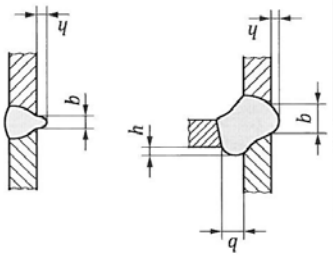
No.	Reference to ISO 6520-1	Imperfection designation	Remarks	t mm	Limits for imperfections for quality levels D	C	B
1 Surface imperfections							
1.1	100	Crack		≥ 0,5	Not permitted	Not permitted	Not permitted
1.2	104	Crater crack		≥ 0,5	Not permitted	Not permitted	Not permitted
1.3	2017	Surface pore	Maximum dimension of a single pore for — butt welds — fillet welds	0,5 to 3	$d \le 0{,}3\ s$ $d \le 0{,}3\ a$	Not permitted	Not permitted
			Maximum dimension of a single pore for — butt welds — fillet welds	> 3	$d \le 0{,}3\ s$, but max. 3 mm $d \le 0{,}3\ a$, but max. 3 mm	$d \le 0{,}2\ s$, but max. 2 mm $d \le 0{,}2\ a$, but max. 2 mm	Not permitted
1.4	2025	End crater pipe		0,5 to 3 > 3	$h \le 0{,}2\ t$ $h \le 0{,}2\ t$, but max. 2 mm	Not permitted $h \le 0{,}1\ t$, but max. 1 mm	Not permitted Not permitted
1.5	401	Lack of fusion (incomplete fusion)	—	≥ 0,5	Not permitted	Not permitted	Not permitted
		Micro lack of fusion	Only detectable by micro examination	≥ 0,5	Permitted	Permitted	Not permitted
1.6	4021	Incomplete root penetration	Only for single side butt welds 	≥ 0,5	Short imperfections: $h \le 0{,}2\ t$ but max. 2 mm	Not permitted	Not permitted

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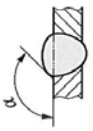
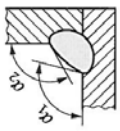
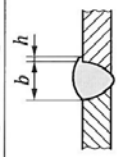
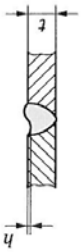
No.	Reference to ISO 6520-1	Imperfection designation	Remarks	t mm	Limits for imperfections for quality levels D	C	B
1.7	5011	Continuous undercut	Smooth transition is required. This is not regarded as a systematic imperfection. 	0,5 to 3	Short imperfections: $h \le 0{,}2\ t$	Short imperfections: $h \le 0{,}1\ t$	Not permitted
	5012	Intermittent undercut		> 3	$h \le 0{,}2\ t$, but max. 1 mm	$h \le 0{,}1\ t$, but max. 0,5 mm	$h \le 0{,}05\ t$, but max. 0,5 mm
1.8	5013	Shrinkage groove	Smooth transition is required. 	0,5 to 3	Short imperfections: $h \le 0{,}2$ mm + $0{,}1\ t$	Short imperfections: $h \le 0{,}1\ t$	Not permitted
				> 3	Short imperfections: $h \le 0{,}2\ t$, but max. 2 mm	Short imperfections: $h \le 0{,}1\ t$, but max.1 mm	Short imperfections: $h \le 0{,}05\ t$, but max. 0,5 mm
1.9	502	Excess weld metal (butt weld)	Smooth transition is required. 	≥ 0,5	$h \le 1$ mm + $0{,}25\ b$, but max. 10 mm	$h \le 1$ mm + $0{,}15\ b$, but max. 7 mm	$h \le 1$ mm + $0{,}1\ b$, but max. 5 mm

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No.	Reference to ISO 6520-1	Imperfection designation	Remarks	t mm	Limits for imperfections for quality levels			
					D	C	B	
1.10	503	Excessive convexity (fillet weld)		≥ 0,5	$h \leq 1$ mm + 0,25 b, but max. 5 mm	$h \leq 1$ mm + 0,15 b, but max. 4 mm	$h \leq 1$ mm + 0,1 b, but max. 3 mm	
1.11	504	Excess penetration		0,5 to 3 > 3	$h \leq 1$ mm + 0,6 b $h \leq 1$ mm + 1,0 b, but max. 5 mm	$h \leq 1$ mm + 0,3 b $h \leq 1$ mm + 0,6 b, but max. 4 mm	$h \leq 1$ mm + 0,1 b $h \leq 1$ mm + 0,2 b, but max. 3 mm	

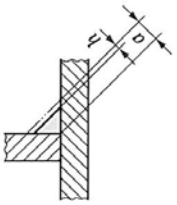
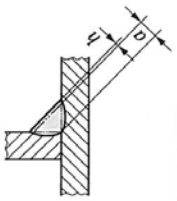
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ISO 5817:2014(E)

No.	Reference to ISO 6520-1	Imperfection designation	Remarks	t mm	Limits for imperfections for quality levels D	C	B
1.12	505	Incorrect weld toe	— butt welds 	≥ 0,5	$\alpha \geq 90°$	$\alpha \geq 110°$	$\alpha \geq 150°$
			— fillet welds  $\alpha_1 \geq \alpha$ and $\alpha_2 \geq \alpha$	≥ 0,5	$\alpha \geq 90°$	$\alpha \geq 100°$	$\alpha \geq 110°$
1.13	506	Overlap		≥ 0,5	$h \leq 0{,}2\,b$	Not permitted	Not permitted
1.14	509	Sagging	Smooth transition is required	0,5 to 3	Short imperfections: $h \leq 0{,}25\,t$	Short imperfections: $h \leq 0{,}1\,t$	Not permitted
	511	Incompletely filled groove		> 3	Short imperfections: $h \leq 0{,}25\,t$, but max. 2 mm	Short imperfections: $h \leq 0{,}1\,t$, but max. 1 mm	Short imperfections: $h \leq 0{,}05\,t$, but max. 0,5 mm

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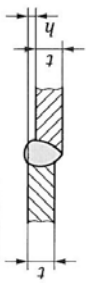
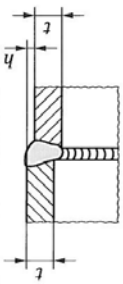
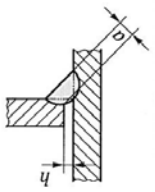
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No.	Reference to ISO 6520-1	Imperfection designation	Remarks	t mm	Limits for imperfections for quality levels			
					D	C	B	
1.18	516	Root porosity	Spongy formation at the root of a weld due to bubbling of the weld metal at the moment of solidification (e. g. lack of gas backing)	≥ 0,5	Locally permitted	Not permitted	Not permitted	
1.19	517	Poor restart	—	≥ 0,5	Permitted The limit depends on the type of imperfection occurred due to restart.	Not permitted	Not permitted	
1.20	5213	Insufficient throat thickness	Not applicable to processes with proof of greater depth of penetration 	0,5 to 3	Short imperfections: $h \leq 0{,}2$ mm + 0,1 a	Short imperfections: $h \leq 0{,}2$ mm	Not permitted	
				> 3	Short imperfections: $h \leq 0{,}3$ mm + 0,1 a, but max. 2 mm	Short imperfections: $h \leq 0{,}3$ mm + 0,1 a, but max. 1 mm	Not permitted	
1.21	5214	Excessive throat thickness	The actual throat thickness of the fillet weld is too large. 	≥ 0,5	Permitted	$h \leq 1$ mm + 0,2 a, but max. 4 mm	$h \leq 1$ mm + 0,15 a, but max. 3 mm	

10

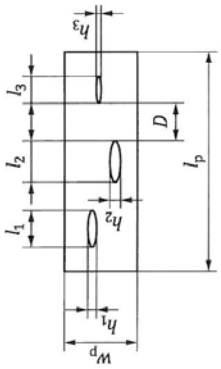
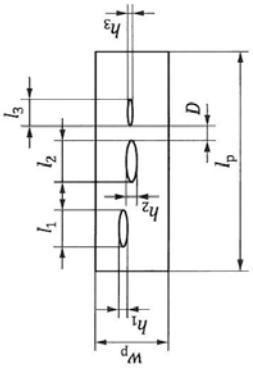
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No.	Reference to ISO 6520-1	Imperfection designation	Remarks	t mm	Limits for imperfections for quality levels D	C	B
3 Imperfections in joint geometry							
3.1	507	Linear misalignment	The limits relate to deviations from the correct position. Unless otherwise specified, the correct position is that when the centrelines coincide (see also Clause 1). t refers to the smaller thickness.				
	5071	Linear misalignment between plates	 Plates and longitudinal welds	0,5 to 3	$h \leq 0{,}2\ \text{mm} + 0{,}25\ t$	$h \leq 0{,}2\ \text{mm} + 0{,}15\ t$	$h \leq 0{,}2\ \text{mm} + 0{,}1\ t$
				> 3	$h \leq 0{,}25\ t$ but max. 5 mm	$h \leq 0{,}15\ t$, but max. 4 mm	$h \leq 0{,}1\ t$, but max. 3 mm
	5072	Transversely circular welds at cylindrical hollow sections	 Circumferential welds	≥ 0,5	$h \leq 0{,}5\ t$, but max. 4 mm	$h \leq 0{,}5\ t$, but max. 3 mm	$h \leq 0{,}5\ t$, but max. 2 mm
3.2	617	Incorrect root gap for fillet welds	Gap between the parts to be joined. Gaps exceeding the appropriate limit may, in certain cases, be compensated for by a corresponding increase in the throat thickness. 	0,5 to 3	$h \leq 0{,}5\ \text{mm} + 0{,}1\ a$	$h \leq 0{,}3\ \text{mm} + 0{,}1\ a$	$h \leq 0{,}2\ \text{mm} + 0{,}1\ a$
				> 3	$h \leq 1\ \text{mm} + 0{,}3\ a$, but max. 4 mm	$h \leq 0{,}5\ \text{mm} + 0{,}2\ a$, but max. 3 mm	$h \leq 0{,}5\ \text{mm} + 0{,}1\ a$, but max. 2 mm

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No.	Reference to ISO 6520-1	Imperfection designation	Remarks	t mm	Limits for imperfections for quality levels		
					D	C	B
4.2	None	Projected or cross-sectional area in longitudinal direction	Case 1 ($D > l_3$)  $h_1 \times l_1 + h_2 \times l_2 + h_3 \times l_3 = \Sigma h \times l$ Case 2 ($D < l_3$)  $h_1 \times l_1 + h_2 \times l_2 + \left(\frac{h_2 + h_3}{2}\right) \times D + h_3 \times l_3 = \Sigma h \times l$ The sum of the areas $\Sigma h \times l$ shall be calculated as a percentage to the evaluation area $l_p \times w_p$ (case 1). If D is smaller than the shorter length of one of the neighbouring imperfections, the full connection of the two imperfections shall be applied to the sum of imperfections (case 2). See also Annex A for information.	≥ 0,5	$\Sigma h \times l \leq 16\ \%$	$\Sigma h \times l \leq 8\ \%$	$\Sigma h \times l \leq 4\ \%$

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1 **APPENDIX 3 – Reference ASW D1.1 Visual Inspection Acceptance Criteria**

sured with suitable gages. Visual inspection for cracks in welds and base metal and other discontinuities should be aided by a strong light, magnifiers, or such other devices as may be found helpful.

6.5.4 Inspector Identification of Inspections Performed. Inspectors shall identify with a distinguishing mark or other recording methods all parts or joints that they have inspected and accepted. Any recording method which is mutually agreeable may be used. Die stamping of cyclically loaded members without the approval of the Engineer shall be prohibited.

6.5.5 Maintenance of Records. The Inspector shall keep a record of qualifications of all welders, welding operators, and tack welders; all WPS qualifications or other tests that are made; and such other information as may be required.

Part B
Contractor Responsibilities

6.6 Obligations of the Contractor

6.6.1 Contractor Responsibilities. The Contractor shall be responsible for visual inspection and necessary correction of all deficiencies in materials and workmanship in conformance with the requirements of this code.

6.6.2 Inspector Requests. The Contractor shall comply with all requests of the Inspector(s) to correct deficiencies in materials and workmanship as provided in the contract documents.

6.6.3 Engineering Judgment. In the event that faulty welding, or its removal for rewelding, damages the base metal so that in the judgment of the Engineer its retention is not in conformance with the intent of the contract documents, the Contractor shall remove and replace the damaged base metal or shall compensate for the deficiency in a manner approved by the Engineer.

6.6.4 Specified NDT Other than Visual. When NDT other than visual inspection is specified in the information furnished to bidders, it shall be the Contractor's responsibility to ensure that all specified welds shall meet the quality requirements of Clause 6, Part C or Clause 9, Part F for tubulars, whichever is applicable.

6.6.5 Nonspecified NDT Other than Visual. If NDT other than visual inspection is not specified in the original contract agreement but is subsequently requested by the Owner, the Contractor shall perform any requested testing or shall allow any testing to be performed in conformance with 6.14. The Owner shall be responsible for all associated costs including handling, surface preparation, NDT, and repair of discontinuities other than those described in 6.9, whichever is applicable, at rates mutually agreeable between Owner and Contractor. However, if such testing should disclose an attempt to defraud or gross nonconformance to this code, repair work shall be done at the Contractor's expense.

Part C
Acceptance Criteria

6.7 Scope

Acceptance criteria for visual and NDT inspection of statically and cyclically loaded nontubular connections are described in Part C. The extent of examination and the acceptance criteria shall be specified in the contract documents in the information furnished to the bidder.

6.8 Engineer's Approval for Alternate Acceptance Criteria

The fundamental premise of the code is to provide general stipulations applicable to most situations. Acceptance criteria for production welds different from those described in the code may be used for a particular application, provided they are suitably documented by the proposer and approved by the Engineer. These alternate acceptance criteria may be based upon evaluation of suitability for service using past experience, experimental evidence or engineering analysis considering material type, service load effects, and environmental factors.

6.9 Visual Inspection

All welds shall be visually inspected and shall be acceptable if the criteria of Table 6.1, or Table 9.16 (if tubular) are satisfied.

6.10 Penetrant Testing (PT) and Magnetic Particle Testing (MT)

Welds that are subject to PT and MT, in addition to visual inspection, shall be evaluated on the basis of the acceptance criteria for visual inspection. The testing shall be performed in conformance with 6.14.4 or 6.14.5, whichever is applicable.

Table 6.1
Visual Inspection Acceptance Criteria (see 6.9)

Discontinuity Category and Inspection Criteria	Statically Loaded Nontubular Connections	Cyclically Loaded Nontubular Connections
(1) Crack Prohibition Any crack shall be unacceptable, regardless of size or location.	X	X
(2) Weld/Base Metal Fusion Complete fusion shall exist between adjacent layers of weld metal and between weld metal and base metal.	X	X
(3) Crater Cross Section All craters shall be filled to provide the specified weld size, except for the ends of intermittent fillet welds outside of their effective length.	X	X
(4) Weld Profiles Weld profiles shall be in conformance with 5.23.	X	X
(5) Time of Inspection Visual inspection of welds in all steels may begin immediately after the completed welds have cooled to ambient temperature. Acceptance criteria for ASTM A514, A517, and A709 Grade HPS 100W [HPS 690W] steels shall be based on visual inspection performed not less than 48 hours after completion of the weld.	X	X
(6) Undersized Welds The size of a fillet weld in any continuous weld may be less than the specified nominal size (L) without correction by the following amounts (U): L, specified nominal weld size, in [mm] / U, allowable decrease from L, in [mm] ≤ 3/16 [5] / ≤ 1/16 [2] 1/4 [6] / ≤ 3/32 [2.5] ≥ 5/16 [8] / ≤ 1/8 [3] In all cases, the undersize portion of the weld shall not exceed 10% of the weld length. On web-to-flange welds on girders, underrun shall be prohibited at the ends for a length equal to twice the width of the flange.	X	X
(7) Undercut (A) For material less than 1 in [25 mm] thick, undercut shall not exceed 1/32 in [1 mm], with the following exception: undercut shall not exceed 1/16 in [2 mm] for any accumulated length up to 2 in [50 mm] in any 12 in [300 mm]. For material equal to or greater than 1 in [25 mm] thick, undercut shall not exceed 1/16 in [2 mm] for any length of weld.	X	
(B) In primary members, undercut shall be no more than 0.01 in [0.25 mm] deep when the weld is transverse to tensile stress under any design loading condition. Undercut shall be no more than 1/32 in [1 mm] deep for all other cases.		X
(8) Porosity (A) CJP groove welds in butt joints transverse to the direction of computed tensile stress shall have no visible piping porosity. For all other groove welds and for fillet welds, the sum of the visible piping porosity 1/32 in [1 mm] or greater in diameter shall not exceed 3/8 in [10 mm] in any linear inch of weld and shall not exceed 3/4 in [20 mm] in any 12 in [300 mm] length of weld.	X	
(B) The frequency of piping porosity in fillet welds shall not exceed one in each 4 in [100 mm] of weld length and the maximum diameter shall not exceed 3/32 in [2.5 mm]. Exception: for fillet welds connecting stiffeners to web, the sum of the diameters of piping porosity shall not exceed 3/8 in [10 mm] in any linear inch of weld and shall not exceed 3/4 in [20 mm] in any 12 in [300 mm] length of weld.		X
(C) CJP groove welds in butt joints transverse to the direction of computed tensile stress shall have no piping porosity. For all other groove welds, the frequency of piping porosity shall not exceed one in 4 in [100 mm] of length and the maximum diameter shall not exceed 3/32 in [2.5 mm].		X

Note: An "X" indicates applicability for the connection type; a shaded area indicates non-applicability.

(3) The backing is transverse to the longitudinal axis of the closed section.

(4) The interruption in the backing does not exceed 1/4 in [6 mm].

(5) The weld with discontinuous backing is not closer than the HSS diameter or major cross section dimension from other types of connections.

(6) The interruption in the backing is not located in the corners.

For statically loaded box columns, discontinuous backing is permitted in the CJP welded corners, at field splices and at connection details. Discontinuous backing is permitted in other closed sections where approved by the Engineer.

NOTE: Commercially available steel backing for pipe and tubing is acceptable, provided there is no evidence of melting on exposed interior surfaces.

9.24 Tolerance of Joint Dimensions

9.24.1 Girth Weld Alignment (Tubular). Abutting parts to be joined by girth welds shall be carefully aligned. No two girth welds shall be located closer than one pipe diameter or 3 ft [1 m], whichever is less. There shall be no more than two girth welds in any 10 ft [3 m] interval of pipe, except as may be agreed upon by the Owner and Contractor. Radial offset of abutting edges of girth seams shall not exceed 0.2t (where t is the thickness of the thinner member) and the maximum allowable shall be 1/4 in [6 mm], provided that any offset exceeding 1/8 in [3 mm] is welded from both sides. However, with the approval of the Engineer, one localized area per girth seam may be offset up to 0.3t with a maximum of 3/8 in [10 mm], provided the localized area is under 8t in length. Filler metal shall be added to this region to provide a 4 to 1 transition and may be added in conjunction with making the weld. Offsets in excess of this shall be corrected as provided in 5.21.3. Longitudinal weld seams of adjoining sections shall be staggered a minimum of 90°, unless closer spacing is agreed upon by the Owner and fabricator.

9.24.2 Groove Dimensions

9.24.2.1 Tubular Cross-Sectional Variations. Variation in cross section dimension of groove welded joints, from those shown on the detailed drawings, shall be in conformance with 5.21.4.1 except:

(1) Tolerances for T-, Y-, and K-connections are included in the ranges given in 9.11.2.

(2) The tolerances shown in Table 9.15 apply to CJP tubular groove welds in butt joints, made from one side only, without backing.

Part F
Inspection

9.25 Visual Inspection

All welds shall be visually inspected and shall be acceptable if the criteria in Table 9.16 are satisfied.

9.26 NDT

9.26.1 Scope. Acceptance criteria for inspection of tubular connections are described in Part C of Clause 6, as applicable, and Part F of this clause. The acceptance criteria shall be specified in the contract documents on information furnished to the bidder.

9.26.2 Tubular Connection Requirements. For CJP groove butt joints welded from one side without backing, the entire length of all completed tubular production welds shall be examined by either RT or UT. The acceptance criteria shall conform to 6.12.1 for RT (see Figure 6.1) or 9.27.1 for UT, as applicable.

9.27 UT

9.27.1 Acceptance Criteria for Tubular Connections. Acceptance criteria for UT shall be as provided in contract documents. Class R or Class X, or both, may be incorporated by reference. Amplitude based acceptance criteria as given by 6.13.1 may also be used for groove welds in butt joints in tubing 24 in [600 mm] in diameter and over, provided all relevant provisions of Clause 6, Part F, are followed. However, these amplitude criteria shall not be applied to tubular T, Y-, and K-connections.

9.27.1.1 Class R (Applicable When UT is Used as an Alternate to RT). All indications having one-half (6 dB) or less amplitude than the standard sensitivity level (with due regard for 9.30.6) shall be disregarded. Indications exceeding the disregard level shall be evaluated as follows:

(1) Isolated random spherical reflectors, with 1 in [25 mm] minimum separation up to the standard sensitivity level shall be accepted. Larger reflectors shall be evaluated as linear reflectors.

(2) Aligned spherical reflectors shall be evaluated as linear reflectors.

Table 9.16
Visual Inspection Acceptance Criteria (see 9.25)

Discontinuity Category and Inspection Criteria	Tubular Connections (All Loads)
(1) Crack Prohibition Any crack shall be unacceptable, regardless of size or location.	X
(2) Weld/Base Metal Fusion Complete fusion shall exist between adjacent layers of weld metal and between weld metal and base metal.	X
(3) Crater Cross Section All craters shall be filled to provide the specified weld size, except for the ends of intermittent fillet welds outside of their effective length.	X
(4) Weld Profiles Weld profiles shall be in conformance with 5.23.	X
(5) Time of Inspection Visual inspection of welds in all steels may begin immediately after the completed welds have cooled to ambient temperature. Acceptance criteria for ASTM A514, A517, and A709 Grade HPS 100W [HPS 690W] steels shall be based on visual inspection performed not less than 48 hours after completion of the weld.	X
(6) Undersized Welds The size of a fillet weld in any continuous weld may be less than the specified nominal size (L) without correction by the following amounts (U): L, specified nominal weld size, in [mm] / U, allowable decrease from L, in [mm] ≤ 3/16 [5] / ≤ 1/16 [2] 1/4 [6] / ≤ 3/32 [2.5] ≥ 5/16 [8] / ≤ 1/8 [3] In all cases, the undersize portion of the weld shall not exceed 10% of the weld length. On web-to-flange welds on girders, underrun shall be prohibited at the ends for a length equal to twice the width of the flange.	X
(7) Undercut (A) For material less than 1 in [25 mm] thick, undercut shall not exceed 1/32 in [1 mm], with the following exception: undercut shall not exceed 1/16 in [2 mm] for any accumulated length up to 2 in [50 mm] in any 12 in [300 mm]. For material equal to or greater than 1 in [25 mm] thick, undercut shall not exceed 1/16 in [2 mm] for any length of weld.	
(B) In primary members, undercut shall be no more than 0.01 in [0.25 mm] deep when the weld is transverse to tensile stress under any design loading condition. Undercut shall be no more than 1/32 in [1 mm] deep for all other cases.	X
(8) Porosity (A) CJP groove welds in butt joints transverse to the direction of computed tensile stress shall have no visible piping porosity. For all other groove welds and for fillet welds, the sum of the visible piping porosity 1/32 in [1 mm] or greater in diameter shall not exceed 3/8 in [10 mm] in any linear inch of weld and shall not exceed 3/4 in [20 mm] in any 12 in [300 mm] length of weld.	
(B) The frequency of piping porosity in fillet welds shall not exceed one in each 4 in [100 mm] of weld length and the maximum diameter shall not exceed 3/32 in [2.5 mm]. Exception: for fillet welds connecting stiffeners to web, the sum of the diameters of piping porosity shall not exceed 3/8 in [10 mm] in any linear inch of weld and shall not exceed 3/4 in [20 mm] in any 12 in [300 mm] length of weld.	X
(C) CJP groove welds in butt joints transverse to the direction of computed tensile stress shall have no piping porosity. For all other groove welds, the frequency of piping porosity shall not exceed one in 4 in [100 mm] of length and the maximum diameter shall not exceed 3/32 in [2.5 mm].	X

Note: An "X" indicates applicability for the connection type; a shaded area indicates non-applicability.